

Q3LOCK submission-preparation guide

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Purpose and honest status

This package is prepared at the author's request without a prior external reviewer. Independent mathematical and specialist novelty review have NOT BEEN PERFORMED. That absence does not prevent preparation of this package, but is not represented as scientific acceptance. The main statement remains conditional, with its exact imported hypotheses, limit orders and parameter conditions visible. Repository result R-497 remains T0 and non-claim-bearing.

Manuscript and contribution

Phase Coexistence in an Eight-Component Q3-Locked Quantum Anharmonic Crystal studies a three-dimensional fixed-spacing lattice with eight real components and a non-radial quartic locking interaction on the internal cube. Seven proof blocks connect finite-volume heat traces, thermodynamic pressure, tempered DLR states, continuous-loop association, spatial reflection positivity and infrared bounds, a collective lower bound, and a strict source cusp with two parity-related DLR states under an explicit sufficient regime.

The candidate model-specific contribution is the positive-locking collective bound and its explicit loop/DLR composition. Standard inequalities, the threshold's functional form and permission for non-radial finite FSS priors are not claimed as new. No novelty or priority certificate is supplied.

Suggested reading order

Read the abstract and Section 1 for scope, Section 2 for the exact model, Section 9 for the conditional main theorem and sufficient regime, and Section 10 for comparisons. The full intervening proofs make the source factors, cutoff ordering and uniform estimates inspectable. Read internal-review.md and the supplement's source and applicability ledgers alongside the proofs. The A1-A23 questions remain available even though no external reviewer could be arranged. Blank review responses are not acceptance.

Nonclaims

The paper does not classify all parameters or the full DLR simplex. It does not claim that the two states are pure, extremal or clustering. It asserts no ground-state gap, infinite-volume real-time KMS construction, continuum limit, physical vacuum, cosmology, Yang-Mills result or closure of any TECT sector.

What is frozen and what is reproduced

The submission TeX is derived from the unchanged v0.1.38 research source by presentation/status changes only. The entire scientific body and bibliography are compared for equality. The manifest identifies all delivered bytes. The archive includes a bounded, hash-checked research snapshot that is actually replayed after extraction; numerical and finite symbolic checks do not prove the infinite-dimensional conclusions. No third-party source PDFs, Git metadata, credentials or private operational records are included.

Files and verification

manuscript.pdf is the reading copy; manuscript.tex is the editable source. The supplement provides theorem-applicability, literature and audit records. internal-review.md records this task's bounded AI rereview. build-report.json, render-review.json and MANIFEST.json identify the tested content and layout. submission-cover-letter.txt is an unsent draft with a substantive contribution statement, not a placeholder awaiting a novelty referee.

After extracting the ZIP, run "python verify-package.py" for integrity. From research-snapshot run "python -X utf8 verification/scripts/q3lock_manuscript_integrated_replay.py --check" with Python 3.12 and SymPy 1.14.0. README.md gives build details. Use the checked read-only command; do not execute historical result writers over the snapshot.

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